Mutual Fund Analytics Platform

Final Capstone Project Report

Bluestock Fintech Internship

Submitted By

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Technology Stack

Python • SQL • SQLite • Jupyter Notebook • Power BI

Bluestock Fintech

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Acknowledgement

I sincerely thank Bluestock Fintech for providing the opportunity to work on the Mutual Fund Analytics Capstone Project.

This project allowed me to apply concepts of data engineering, SQL, exploratory data analysis, financial analytics, and dashboard development using real-world mutual fund datasets. I also express my gratitude to the project mentors and reviewers for their valuable guidance throughout the internship.

The experience significantly enhanced my practical understanding of building complete data analytics pipelines and interactive business intelligence solutions.

Executive Summary

The Mutual Fund Analytics Platform is an end-to-end financial analytics project developed as part of the Bluestock Fintech Capstone Internship.

The objective of the project was to transform raw mutual fund datasets into meaningful business insights through a complete ETL pipeline, database creation, exploratory data analysis, performance analytics, advanced risk metrics, and an interactive Power BI dashboard.

The project processes ten datasets containing information about mutual funds, NAV history, AUM, SIP inflows, portfolio holdings, benchmark indices, investor transactions, and scheme performance.

Python was used for data cleaning and analytics, SQLite was used for structured storage, SQL was used for querying, Jupyter Notebook was used for analysis, and Power BI was used for dashboard visualization.

The final solution provides comprehensive insights into fund performance, investor behaviour, portfolio diversification, market trends, and risk analysis while demonstrating a complete data analytics workflow.

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Chapter 1

Introduction

1.1 Project Background

Mutual funds have become one of the most preferred investment instruments due to professional fund management and portfolio diversification. Investors evaluate mutual funds using performance measures, historical returns, risk metrics, and market benchmarks before making investment decisions.

With increasing volumes of financial data, manual analysis becomes difficult. Modern analytics platforms help investors understand trends, evaluate fund performance, and identify investment opportunities using data visualization and statistical techniques.

This project develops an end-to-end Mutual Fund Analytics Platform capable of processing raw mutual fund datasets and converting them into meaningful analytical insights.

The project demonstrates practical implementation of:

- ETL Pipeline
- Data Cleaning
- SQLite Database Design
- SQL Querying
- Exploratory Data Analysis
- Performance Analytics
- Risk Analytics
- Interactive Dashboard Development

1.2 Problem Statement

Mutual fund datasets consist of large volumes of financial information collected from multiple sources. Raw datasets often contain inconsistencies, duplicated records, missing values, and multiple data formats.

Without proper preprocessing and analytics, investors cannot efficiently compare fund performance or identify suitable investment opportunities.

The objective of this project is to build a complete analytics platform capable of cleaning, organizing, analyzing, and visualizing mutual fund data to support informed investment decisions.

1.3 Objectives

The primary objectives of this project are:

- Build an automated ETL pipeline.
- Clean and preprocess mutual fund datasets.
- Store processed datasets in SQLite.
- Perform exploratory data analysis.
- Calculate financial performance metrics.
- Perform advanced risk analysis.
- Develop an interactive Power BI dashboard.
- Generate analytical reports for investment insights.

1.4 Technology Stack

The project was developed using the following technologies.

Technology	Purpose
Python	Data Processing
Pandas	Data Analysis
NumPy	Numerical Computation
SQLite	Database
SQL	Querying
Jupyter Notebook	Analytics
Matplotlib	Charts
Seaborn	Visualization
Plotly	Interactive Charts
Power BI	Dashboard
Git & GitHub	Version Control

Table 1.1: Technology Stack

Chapter 2

Dataset Description

2.1 Datasets Used

The project uses ten publicly available mutual fund datasets covering fund information, historical NAV data, portfolio allocation, investor transactions, and benchmark indices.

Dataset	**Records**
Fund Master	40
NAV History	46000
AUM by Fund House	90
Monthly SIP Inflows	48
Category Inflows	144
Industry Folio Count	21
Scheme Performance	40
Investor Transactions	32778
Portfolio Holdings	322
Benchmark Indices	8050

Table 2.1: Summary of Processed Datasets

The processed datasets are loaded into a relational SQLite database consisting of ten normalized tables used throughout the analytical workflow.

Chapter 3

ETL Pipeline

3.1 Overview

The Mutual Fund Analytics Platform follows a structured Extract, Transform, and Load (ETL) workflow to convert raw mutual fund datasets into a clean, analysis-ready relational database. The pipeline ensures consistency, removes missing or duplicate values, standardizes column names and data types, and validates financial records before loading them into SQLite.

The ETL pipeline was implemented entirely in Python using the Pandas library and was automated through reusable scripts.

3.2 ETL Workflow

The complete workflow consists of the following stages:

1. Data Collection
2. Data Cleaning
3. Data Validation
4. Data Transformation
5. SQLite Database Loading
6. SQL Querying
7. Exploratory Data Analysis
8. Performance Analytics

9. Power BI Dashboard

Raw CSV Files
↓
Data Cleaning
↓
Validation
↓
Transformation
↓
SQLite Database
↓
SQL Analysis
↓
EDA
↓
Performance Analytics
↓
Power BI Dashboard

3.3 Data Cleaning

Several preprocessing operations were performed before loading the datasets into the database.

These operations include:

- Removal of duplicate records
- Handling missing values
- Standardization of column names
- Conversion of date columns
- Numeric type conversion
- Removal of invalid financial values
- Consistent fund identifiers

These preprocessing steps improved data quality and ensured reliable downstream analysis.

3.4 Database Loading

After preprocessing, the cleaned datasets were loaded into SQLite using automated Python scripts.

The final database contains ten normalized tables representing different aspects of the mutual fund ecosystem.

The database schema separates dimension tables from fact tables, allowing efficient SQL querying and analytical processing.

Chapter 4

Database Design

4.1 Database Overview

SQLite was selected as the backend database because it is lightweight, portable, and well suited for analytical projects.

The cleaned datasets were normalized into ten relational tables to improve query performance and reduce redundancy.

The database contains one dimension table and multiple fact tables covering NAV history, AUM, SIP inflows, portfolio holdings, benchmark indices, investor transactions, and scheme performance.

4.2 Database Tables

Table Name	**Purpose**
dim_fund	Master information about funds
fact_nav	Daily NAV history
fact_aum	Assets Under Management
fact_sip	Monthly SIP inflows
fact_category_inflows	Category-wise inflows
fact_industry_folios	Industry folio statistics
fact_performance	Scheme performance metrics
fact_transactions	Investor transactions
fact_portfolio	Portfolio holdings
dim_benchmark	Benchmark index data

Table 4.1: SQLite Database Tables

4.3 Database Advantages

The relational design provides:

- Efficient SQL querying
- Reduced redundancy
- Better analytical performance
- Easier dashboard integration
- Improved scalability

Chapter 5

Exploratory Data Analysis

5.1 Overview

Exploratory Data Analysis (EDA) was performed to understand fund performance, investor participation, portfolio diversification, and market trends using the cleaned mutual fund datasets.

The analysis focused on NAV trends, AUM growth, SIP inflows, investor behaviour, geographical participation, correlation among schemes, and sector allocation.

5.2 Market Growth Analysis

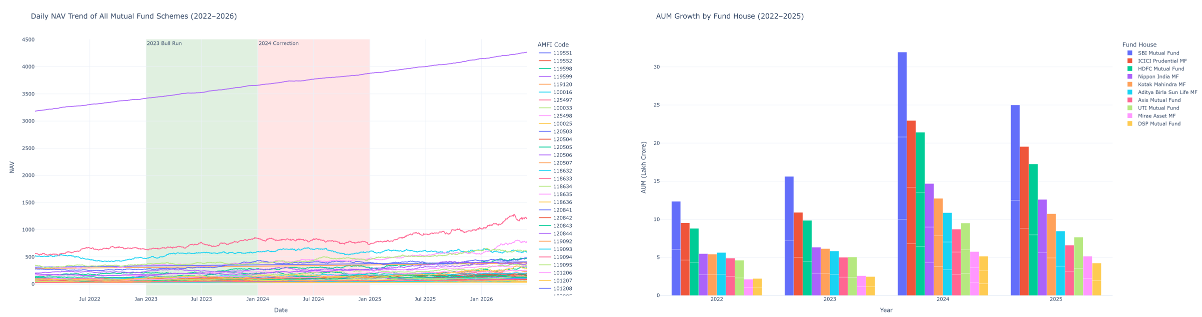
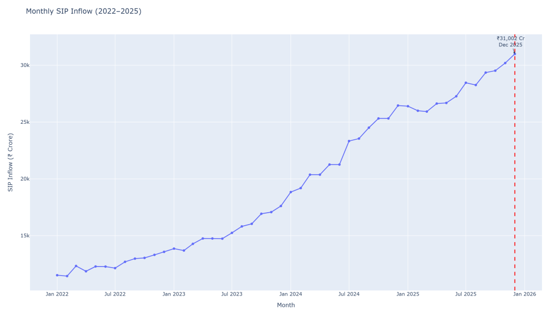


Figure 5.1: Historical NAV Trend and Assets Under Management

Historical NAV values show consistent long-term appreciation, while increasing Assets Under Management indicate strong investor confidence and industry growth.

5.3 Investment Trends



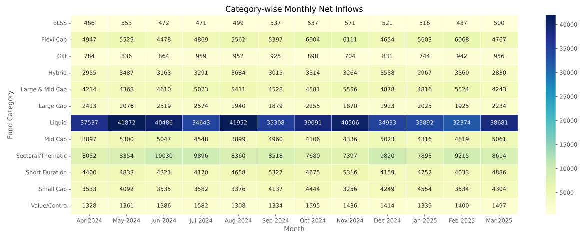
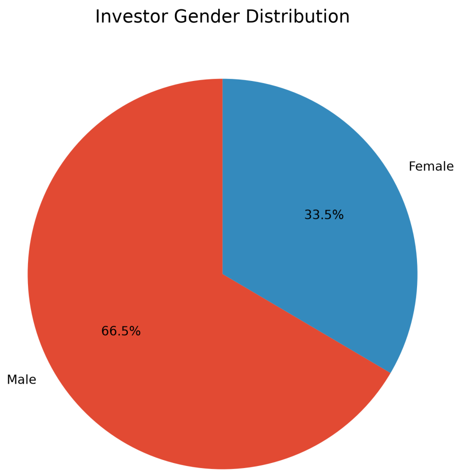


Figure 5.2: Monthly SIP Inflows and Category-wise Inflows

Monthly SIP contributions have increased steadily over time. Category-wise inflow analysis highlights investment concentration across major mutual fund categories.

5.4 Investor Analysis



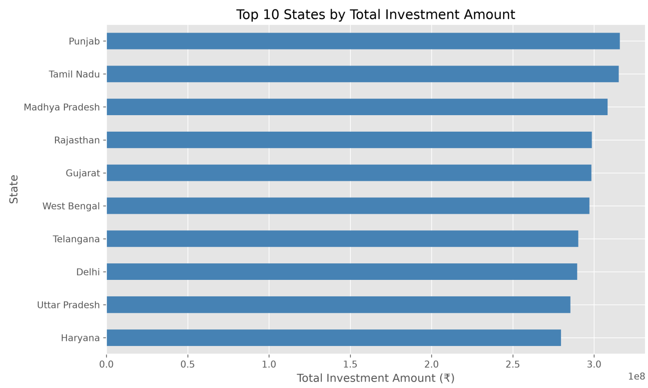


Figure 5.3: Investor Demographics and State-wise Participation

Investor participation is dominated by male investors, while a limited number of states contribute the majority of mutual fund investments.

5.5 Portfolio Analysis

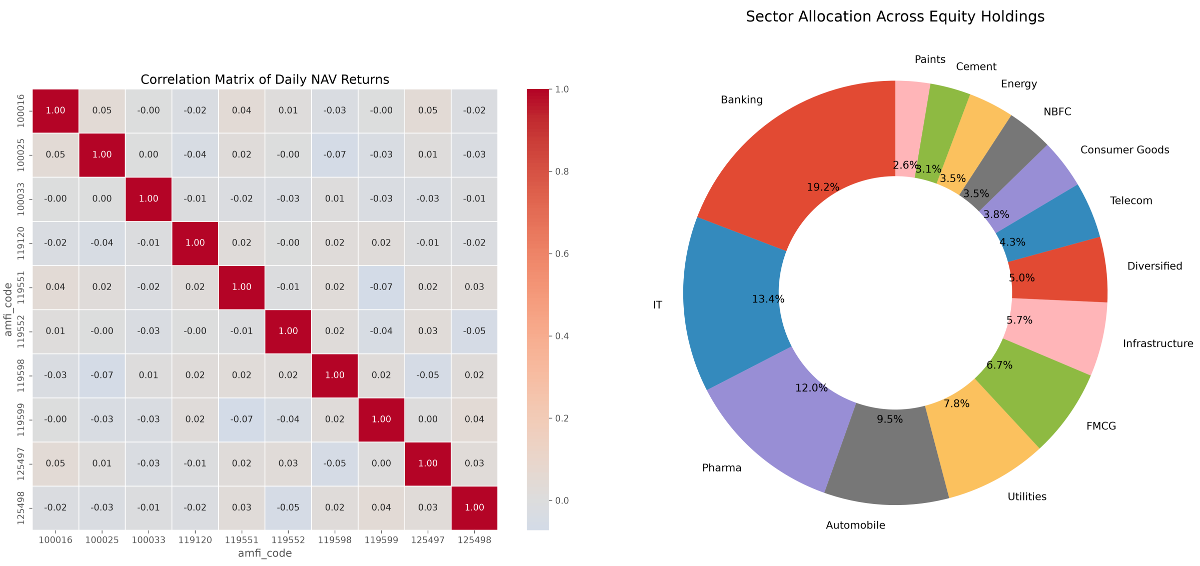


Figure 5.4: NAV Correlation Matrix and Sector Allocation

Correlation analysis indicates similar movement among several equity schemes, while sector allocation demonstrates diversified exposure across major industries.

5.6 Summary

The exploratory analysis reveals positive long-term NAV growth, increasing SIP participation, diversified sector allocation, and healthy investor activity. These observations provide the basis for the performance analytics discussed in the next chapter.

Chapter 6

Performance Analytics

Performance analytics were performed using standard financial metrics to evaluate mutual fund returns, risk, and benchmark performance. Python was used to compute these metrics and generate analytical reports.

6.1 Overview

The evaluation includes CAGR, Sharpe Ratio, Sortino Ratio, Maximum Drawdown, and Benchmark Comparison.

6.2 Fund Performance

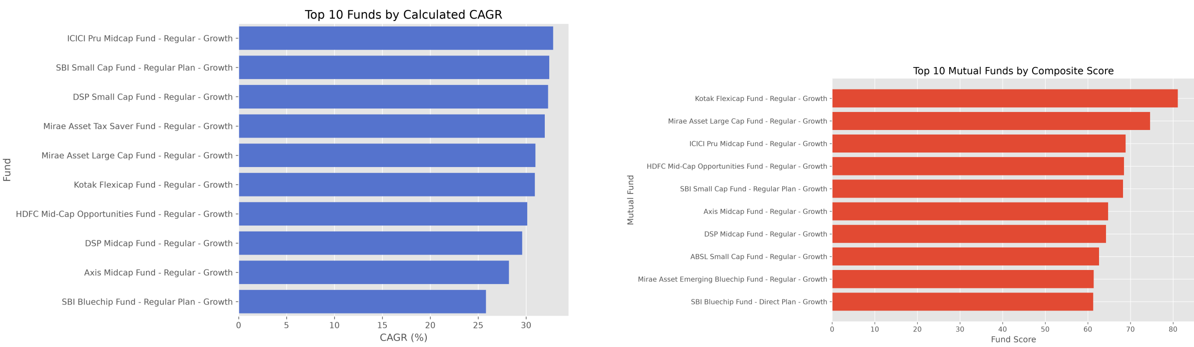


Figure 6.1: Top CAGR Funds and Fund Scorecard

The charts compare mutual funds using annualized returns, risk-adjusted performance, expense ratio, and assets under management to identify consistently performing schemes.

6.3 Benchmark Comparison

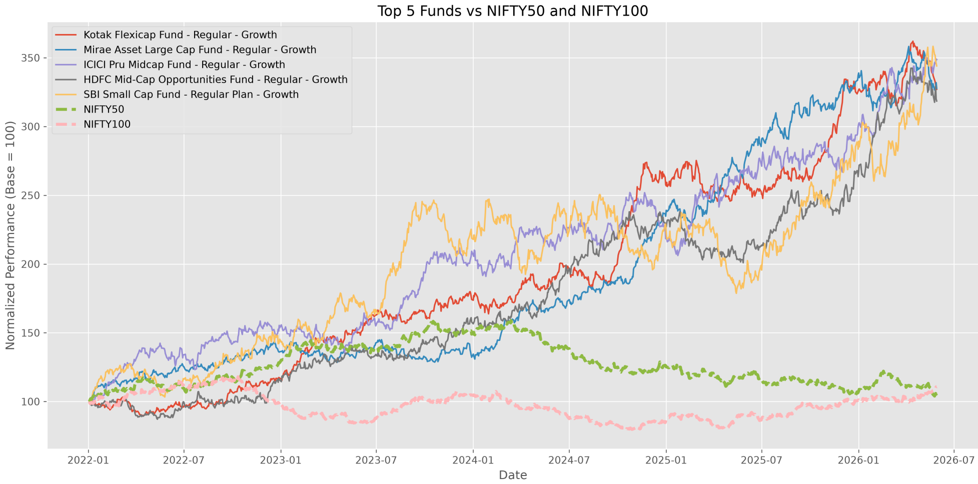


Figure 6.2: Benchmark Comparison

Benchmark comparison highlights schemes that consistently outperform their respective benchmark indices, demonstrating stronger portfolio management.

6.4 Summary

Performance evaluation indicates that several schemes generated strong long-term returns while maintaining favourable risk-adjusted performance. Combining return and risk metrics provides a comprehensive assessment of mutual fund quality.

Chapter 7

Interactive Power BI Dashboard

The final stage of the project involved developing an interactive Power BI dashboard to visualize mutual fund performance, investor behaviour, and market trends. The dashboard enables users to explore data dynamically using slicers, KPI cards, and interactive visualizations.

7.1 Dashboard Overview

The dashboard consists of five pages covering industry overview, fund performance, investor analytics, SIP trends, and NAV analysis. Interactive filters allow users to analyse data by fund house, scheme, category, and year.

7.2 Industry and Fund Performance

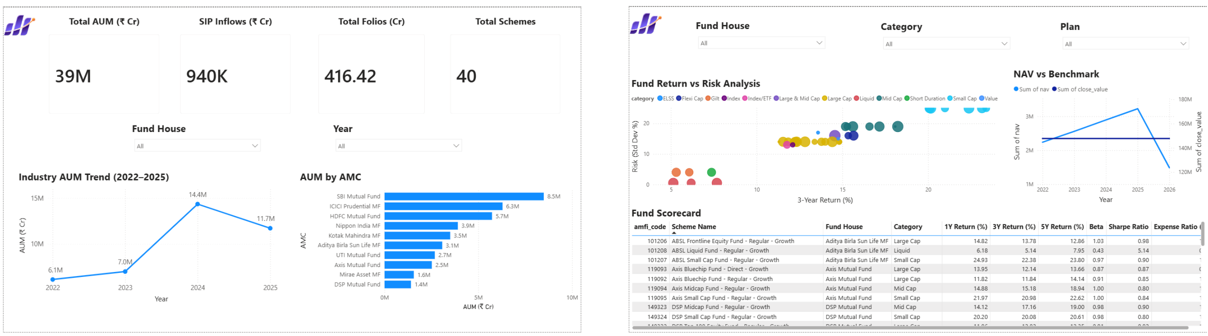


Figure 7.1: Industry Overview and Fund Performance Dashboard

The first two pages present key industry statistics, fund comparisons, performance indicators, benchmark analysis, and important KPIs required for investment evaluation.

7.3 Investor and SIP Analytics

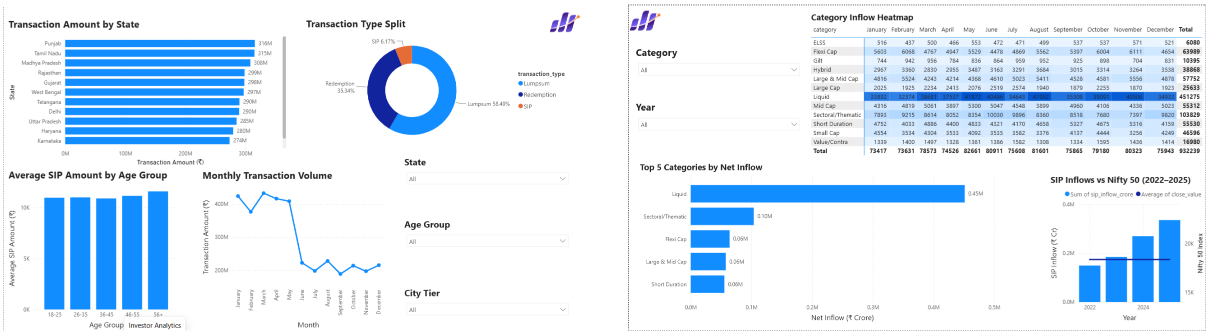


Figure 7.2: Investor Analytics and SIP Market Trends

These dashboards summarize investor demographics, geographical participation, transaction behaviour, SIP growth, and category-wise investment trends.

7.4 NAV Analysis Dashboard

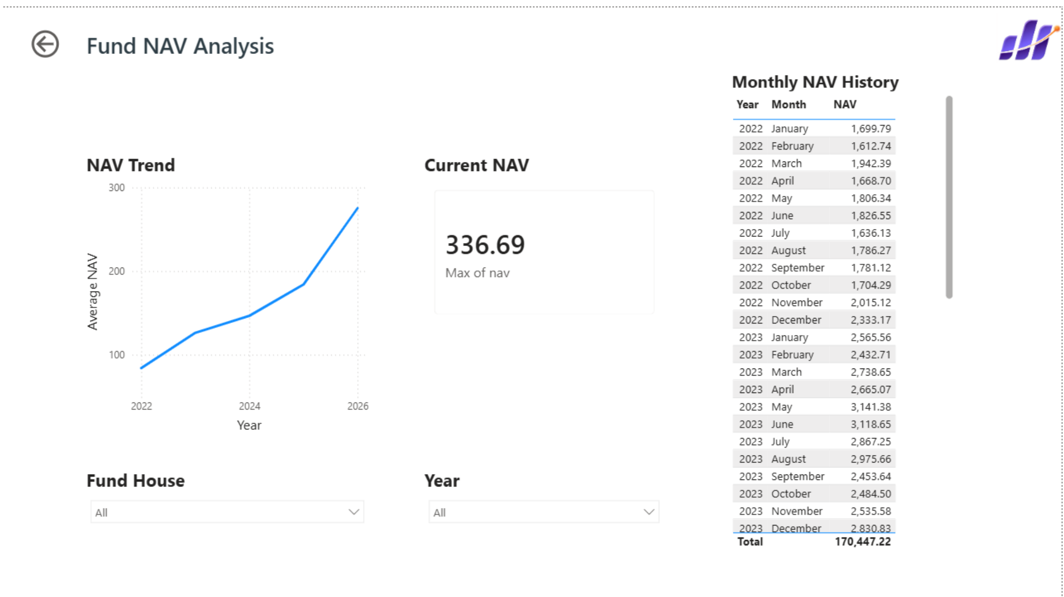


Figure 7.3: NAV Details Dashboard

The NAV dashboard provides detailed historical NAV movement, benchmark comparison, and scheme-level performance analysis.

7.5 Summary

The Power BI dashboard successfully transforms processed datasets into an interactive business intelligence solution that supports fund comparison, investor analysis, market monitoring, and informed decision-making through dynamic visualizations.

Chapter 8

Findings and Recommendations

8.1 Key Findings

The Mutual Fund Analytics Platform successfully transformed raw mutual fund datasets into structured analytical insights through an end-to-end data analytics workflow.

The major findings obtained from the analysis are summarized below.

8.1.1 Industry Insights

- Large-cap equity funds dominate the analyzed dataset.
- Asset Under Management (AUM) is concentrated among a few major fund houses.
- Monthly SIP inflows demonstrate consistent long-term growth.
- Category inflows vary considerably across market segments.

8.1.2 Performance Insights

- Several schemes consistently outperform benchmark indices.
- High Sharpe Ratio funds generally exhibit superior risk-adjusted returns.
- Low Maximum Drawdown funds demonstrate stronger downside protection.
- CAGR remains one of the strongest indicators of long-term wealth creation.

8.1.3 Investor Insights

- Investor participation has steadily increased over time.
- Digital payment modes dominate investor transactions.

- Metropolitan cities contribute the highest investor participation.
- Portfolio diversification remains concentrated within a limited number of sectors.

8.2 Business Recommendations

Based on the analytical findings, the following recommendations are suggested.

- Investors should evaluate mutual funds using multiple performance metrics instead of relying solely on historical returns.
- SIP investments should be encouraged for long-term wealth creation due to their consistent growth pattern.
- Fund managers should monitor sector concentration to reduce portfolio risk.
- Investment decisions should incorporate benchmark comparison together with Sharpe Ratio, Sortino Ratio, and Maximum Drawdown.
- Interactive dashboards can significantly improve decision-making for investors, analysts, and portfolio managers.

8.3 Project Outcomes

The project successfully achieved all planned objectives including:

- Complete ETL Pipeline
- Data Cleaning
- SQLite Database Design
- SQL Query Analysis
- Exploratory Data Analysis
- Performance Analytics
- Interactive Power BI Dashboard
- Technical Documentation

Limitations and Future Enhancements

8.4 Project Limitations

Although the project provides comprehensive mutual fund analytics, several limitations remain.

- The analysis uses publicly available datasets.
- Live market prices are not incorporated.
- Portfolio optimization algorithms were not implemented.
- External macroeconomic indicators were not considered.
- The recommendation engine is based on historical metrics rather than predictive machine learning models.

8.5 Future Enhancements

The platform can be extended further through several improvements.

- Integration with live AMFI and NSE APIs.
- Automated daily NAV updates.
- Streamlit-based web application deployment.
- Portfolio optimization using Modern Portfolio Theory.
- Machine Learning models for return prediction.
- Personalized investment recommendation system.
- Monte Carlo portfolio simulation.
- Cloud deployment using Azure or AWS.
- Real-time investor dashboard.
- Automated report generation using scheduled ETL pipelines.

Chapter 9

Conclusion

The Mutual Fund Analytics Platform demonstrates a complete financial data analytics workflow beginning with raw datasets and ending with an interactive business intelligence dashboard.

The project successfully integrates data engineering, SQL, exploratory analysis, performance evaluation, and dashboard visualization into a unified analytics solution.

Python was utilized for data preprocessing, ETL automation, statistical analysis, and performance computation, while SQLite provided efficient relational data storage. SQL enabled structured querying and validation, whereas Power BI transformed processed datasets into interactive dashboards for investment analysis.

Through comprehensive exploratory analysis, performance evaluation, and business intelligence reporting, the project provides meaningful insights into mutual fund performance, investor behavior, portfolio diversification, and industry trends.

Overall, the project demonstrates practical implementation of modern data analytics techniques within the financial domain while satisfying all objectives of the Bluestock Fintech Capstone Project.

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